

The Isotropic Radiation

Ground State

and the *Traction Engine*

Why H^s moves: the budget, the partition, the log-frequency carrier, and the phase trajectory — derived from thirty years of measured acoustic work, written as a single formula, with the full mathematical apparatus.

MASTER STANDARD

Working flagship paper · draft v2.1

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Scope discipline

This is a H^s flagship paper. It is **not** part of the CoDaWork 2026 conference package and falls **outside** the Pre-Conference Lockdown window (2026-05-12 → 2026-06-06). It is a separate post-conference project that can be drafted, refined, and committed during the lockdown without touching any locked surface.

Citation policy

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Preface — why this document exists, and why now

H^s works. It worked the first time it was tried on energy-mix data, it worked on the geochemistry corpus, it worked on the macro-economic compositions, it worked on the eleven domains and one hundred and one datasets of the 2026-05 validation suite. Reviewers ask why, and the published reply — *"it inherits from compositional data analysis and adds a quaternion phase layer"* — is true but insufficient. It explains the structure. It does not explain the **confidence**.

This document supplies the missing explanation. The confidence is not philosophical. It is empirical. The H^s framework is the formal generalization of a thirty-year body of acoustic engineering practice conducted at the **Binaural Test Lab** — a sound-controlled professional laboratory operated through a private industrial sponsorship with parallel deployments in Ottawa and Monaco — where the same compositional structure has been validated against measured sound fields over working hours, under varying environmental conditions, across real listening positions, with real human listeners, on real loudspeaker hardware, in real rooms. The mathematics did not arrive first and find an application. The acoustic measurements arrived first and forced the mathematics.

Three properties of the original acoustic problem turn out to be the three properties that make H^s a traction engine — a framework that moves, not one that holds still:

- 1. A physical ground state fixes the total.** The unit-sphere radiation budget — 6.02 dB across the $4\pi \rightarrow 2\pi$ baffle-step transition — is a measured invariant, not a normalization convention. Closure is enforced by acoustic physics, not chosen for mathematical convenience.
- 2. The partition lives on the simplex.** The total is distributed across cabinet dimensions in DADC, across ERB bands × drivers in HCI-AUDIO, across compositional carriers in the general case. The simplex is where the parts live. It is the same simplex regardless of the domain.
- 3. Time enters the simplex through the log-frequency axis.** This is the part that has never before been written down in one place. The geometric-frequency association — the fact that the partition is naturally indexed by log-spaced carriers — couples directly to group delay, which couples directly to phase rotation on the three-sphere S^3 . The simplex acquires *motion* because each carrier has a log-frequency position and a phase trajectory. The static partition becomes a path. The path is traction.

The unified formula below carries all four ingredients in a single expression. Around it sit the mathematical foundations that have always supported the acoustic instance and that now make the generalization rigorous: Helmholtz reciprocity (Lemma 3), the Rayleigh-Sommerfeld integral (Lemma 2), Banach fixed-point convergence (Lemma 4), spectral-radius stability of the adaptive feedback (Lemma 5), Statistical Energy Analysis positive-definiteness (Lemma 6), the group-delay-as-rotation identity on S^3 (Lemma 7), and closure invariance under the log-ratio transform (Lemma 8). The chain closes with the master statement of the unified formula (Theorem 1) and its compositional generalization (Theorem 2).

This document is the first time all of it has been written down as a unified, internally-proven statement. It is intended to be the master standard for the chain.

§1 — Symbols and notation

The notation below follows the conventions established in HCI-CNT/handbook/GLOSSARY.md v3.0 §26. Symbols introduced in this document add to that base set; they are listed below with their first-use section number.

Scalars and physical constants

Symbol	Meaning	First use
c	Ground-state radiation budget = $20 \cdot \log_{10}(2) \approx 6.0206$ dB	§2
c_{sound}	Speed of sound in air, $\approx 343 \text{ m s}^{-1}$ at 20 °C	§4
ρ	Air density, $\approx 1.21 \text{ kg m}^{-3}$ at 20 °C, 101.3 kPa	§7 (Lemma 2)
λ	Wavelength, $\lambda = c_{\text{sound}} / f$	§7
k	Wavenumber, $k = 2\pi / \lambda = \omega / c_{\text{sound}}$	§7
ω	Angular frequency, $\omega = 2\pi f$	§5
κ	Evanescent decay constant, $\kappa = \sqrt{(k^2 - \omega^2/c_{\text{sound}}^2)}$ for $ k > \omega/c_{\text{sound}}$	§7

Geometric and partition quantities

Symbol	Meaning	First use
n	Number of partitions in the active instance	§3
dim_i	Physical extent of the i -th partition (acoustic case)	§3
S	Geometric scale, $S = \sum_i \text{dim}_i$	§3
I_s	Reciprocal scale, $I_s = \sum_i (1 / \text{dim}_i)$	§3
D	Dominance ratio, $D = \max(\text{dim}_i) / \min(\text{dim}_i)$	§3
β	Hybrid-regime blend coefficient, $\beta = 2(D - 1.5)$	§3
p_i	Simplex coordinate (portion), $p_i = \text{dim}_i / S$; $\sum p_i = 1$	§3
G_i	Per-partition gain (dB), $G_i = c \cdot p_i$; $\sum G_i = c$	§3

Spectral and phase quantities

Symbol	Meaning	First use
f	Frequency (Hz)	§4
$F_{c,i}$	Geometric cutoff frequency for partition i , $F_{c,i} = 115 / \text{dim}_i$	§4

Symbol	Meaning	First use
$\bar{f}_j$	Geometric centre frequency of band j	§4
$S(f, F_{c,i})$	First-order baffle-step shelf, $1/\sqrt{1 + (F_{c,i}/f)^2}$	§4
$\varphi(f)$	Phase response at frequency f	§5
τ	Group delay (seconds), $\tau = -(1/2\pi) d\varphi/df$	§5
τ_i	Per-partition group delay	§5
$\hat{n}_i$	Unit rotation axis for partition i on S^3	§5
$q(f, t)$	Unit-quaternion phase state at frequency f , time t	§5
i_x, i_y, i_z	Three imaginary units of the quaternion algebra	§5

H^s system quantities

(per HCI-CNT/handbook/GLOSSARY.md §26)

Symbol	Meaning	First use
D	Composition dimension (number of carriers)	§10
T	Number of records (timesteps)	§10
x_i, ρ_i	The i -th carrier's value or share	§10
$\text{clr}_i(t)$	Centred log-ratio coordinate at time t	§7 (Lemma 8)
$\eta(t)$	ILR coordinate vector at time t	§7 (Lemma 8)
$\alpha_j(t)$	Activation Coefficient for carrier j at time t	§11
$\Pi_j(t)$	Power Share for carrier j at time t	§11
$Q(t)$	Quaternion trajectory as a function of time	§11
$S^{(D-1)}$	The $(D-1)$ -simplex	§3
S^3	The 3-sphere $\cong$ unit quaternions $\cong SU(2)$	§5

Operators

Symbol	Meaning	First use
$C(\cdot)$	Closure operator, $C(x) = x / \sum x_i$	§3
$\text{clr}(\cdot)$	Centred log-ratio, $\text{clr}_i(x) = \log(x_i) - (1/D) \sum_j \log(x_j)$	§7 (Lemma 8)
$\text{ilr}(\cdot) = V^T \text{clr}(\cdot)$	Isometric log-ratio (Helmert) with $VV^T = I$	§7 (Lemma 8)
q^*	Quaternion conjugate, $(a, -b, -c, -d)$	§5
Δ	Forward difference, $\Delta f(t) = f(t+1) - f(t)$	§11

Symbol	Meaning	First use
∇^2	Laplacian operator	§7 (Lemma 2)

§2 — The isotropic radiation ground state

Claim. When a point acoustic source is mounted in a rigid finite baffle, the radiation field undergoes a measured transition between two limiting regimes:

Low frequency. Wavelength is large compared to the baffle dimensions. The source radiates into the full 4π steradian sphere — *isotropic radiation* — and the response carries an inverse-square-law roll-off relative to the half-space reference.

High frequency. Wavelength is small compared to the baffle dimensions. The source radiates into the forward 2π steradian half-space and the response is flat relative to the same reference.

The transition between these two regimes is the *baffle step*. Its total magnitude is

$$\Delta L = 20 \cdot \log_{10}(2) \approx 6.0206 \text{ dB.} \quad (1)$$

This is the **isotropic radiation ground state of the loudspeaker problem**. The 6.02 dB figure is the on-axis pressure ratio that results from a *power-conserving* redistribution: the same total radiated acoustic power, integrated over the sphere, transitions from being spread across 4π steradians to being concentrated into 2π . Total power is conserved; on-axis pressure rises by exactly $20 \cdot \log_{10}(2)$. It is not approximate, not a convention, not a model parameter. In calibrated measurement it reproduces to better than 0.05 dB. At the Binaural Test Lab it has been observed to that precision continuously, under varying temperature, humidity, and atmospheric pressure, for more than three decades.

The conservation law that makes the budget interpretation legitimate is the conservation of *total radiated acoustic power*. The 6.02 dB itself is in pressure (amplitude) decibels, but the closure (equation 14) holds because **what is being apportioned is the total radiated power across the partitions, not the on-axis amplitude**. This distinction is the key to §4.2 below, and it determined the crossover topology that ships with every BTL build.

Define

$$c := 20 \cdot \log_{10}(2) = 6.0206 \text{ dB.} \quad (2)$$

The constant **c** is the *ground-state budget*. Everything in what follows is a distribution of, or a phase trajectory around, this budget. It is the analogue of the closure constraint $\sum p_i = 1$ in standard compositional-data analysis (Aitchison 1986; Pawlowsky-Glahn et al. 2015) — but the analogue carries a physical interpretation that compositional data analysis (CoDa) typically lacks: the constant is *the total amount of radiation*, not an arbitrary unit of accountancy.

§2.1 — Why this is the ground state and not just a normalization

In quantum-mechanical language, a *ground state* is the lowest-energy configuration of a system, against which all excitations are measured. The 6.02 dB radiation budget plays exactly this role in acoustic baffle problems:

- Every diffraction correction is a deviation from the isotropic budget.
- Every cabinet-dimension allocation is a partition of the budget.
- Every adaptive feedback step (ADAC) is a return to the budget.

The budget is what the system relaxes to in the absence of forcing. The partitions and portions are how the budget gets *allocated* under physical constraints — driver size, cabinet geometry, listening-position aiming. The closure rule is what *holds the allocation together* as conditions change.

This is the same triple structure that appears in every successful application of H^s to a non-acoustic problem. The energy-mix monitoring work uses electrical-generation share (closure: 100 %) as the budget; the geochemistry work uses major-element oxide fraction (closure: 100 % by weight); the macro-economic work uses GDP share (closure: 100 %). In each case the partition is structural, the closure is physical, and the deviations from the equal-share isotropic reference are the *signals*. The 6.02 dB was the first one, and it was measured before anyone in compositional-data analysis had thought to look for it.

§3 — Partitions and portions on the simplex

The forward map (DADC). Given cabinet dimensions (H, W, D) with $H + W + D = S$, the Dimension-Apportioned Diffraction Correction distributes the ground-state budget c as

$$G_i = c \cdot \text{dim}_i / S \quad \text{for } i \in \{H, W, D\} \quad (3)$$

$$\sum_i G_i = c. \quad (4)$$

Equation (4) is the **closure rule**. It is the same closure rule that appears in every CoDa application; it is enforced here by the physical observation that the total radiation budget is fixed. The simplex coordinate $p_i = \text{dim}_i / S$ is the normalized portion. The actual gain G_i in decibels is the portion multiplied by the ground-state constant c . The closure proof is Lemma 1 in §7.

The BTL measurement. For the canonical BTL rectangular geometry ($H = 0.8$ m, $W = 0.368$ m, $D = 0.33$ m, $S = 1.498$ m):

Dimension	dim_i (m)	$p_i = \text{dim}_i / S$	G_i (dB)
Height	0.8	0.534	3.215
Width	0.368	0.246	1.479
Depth	0.33	0.220	1.326
Total	1.498	1.000	6.020

The closure is exact to four decimal places. It is exact in every well-calibrated BTL measurement *because it is forced by physics*.

The generalization. Replacing physical dimensions with arbitrary compositional carriers and replacing the 6.02 dB ground state with whatever total constraint the application imposes gives the standard CoDa apportionment

$$p_i = x_i / \sum_j x_j, \quad \sum_i p_i = 1. \quad (5)$$

Equation (5) is what one finds in Aitchison (1986) and Pawlowsky-Glahn et al. (2015). Equation (3) is the *acoustic instance* of equation (5). The simplex was already there in the $4\pi \rightarrow 2\pi$ physics; CoDa later supplied the geometry; H^s supplied the dynamics. The order matters historically. The mathematics is the same.

§3.1 — Short, long, and hybrid regimes

The single-formula treatment in equation (3) holds for the **long-dimension regime** ($D = \max(\dim_i)/\min(\dim_i) > 2$). In two adjacent regimes the apportionment changes form:

Short regime ($D < 1.5$). Reciprocal emphasis: $G_i = -c \cdot (1/\dim_i) / I_s$ where $I_s = \sum_j (1/\dim_j)$. Closure remains $\sum G_i = -c$.

Hybrid regime ($1.5 \leq D \leq 2$). Linear blend: $G_i = c \cdot [\beta \cdot \dim_i/S + (1 - \beta) \cdot (1/\dim_i)/I_s]$ with $\beta = 2(D - 1.5)$.

In all three regimes the closure $\sum G_i = \pm c$ holds exactly. The simplex constraint is regime-independent; only the orientation of the apportionment within the simplex changes. This is the acoustic precursor of the **Helmsman family** in CNT (sign / stability / flips / chaos / torque / joint) — a single closure with multiple regime-specific orientations.

§4 — The geometric-frequency association

The cutoff. Each cabinet dimension has an associated cutoff frequency

$$F_{c,i} = 115 / \text{dim}_i \quad (\text{Hz, with dim}_i \text{ in metres}) \quad (6)$$

derived from $c_{\text{sound}} / (2 \cdot \text{dim}_i)$ where $c_{\text{sound}} \approx 343 \text{ m s}^{-1}$ is the speed of sound. For the BTL geometry:

Dimension	$F_{c,i}$ (Hz)
Height	143.75
Width	312.50
Depth	348.48

The transition for each dimension is a first-order baffle-step shelf,

$$S(f, F_{c,i}) = 1 / \sqrt{1 + (F_{c,i} / f)^2}. \quad (7)$$

Why log-frequency, not linear. Two independent facts make the log-frequency axis the natural one:

- 1. Geometric scaling.** Cutoff frequencies inversely related to physical dimensions cluster geometrically: doubling a dimension halves its cutoff. The natural index variable is $\log F$, because differences in $\log F$ correspond to ratios of dimension. The dimensions $H = 0.8$, $W = 0.368$, $D = 0.33$ give cutoffs spaced by ratios 2.18 and 1.12, not by absolute differences.
- 2. Perceptual scaling.** Human auditory perception is logarithmic in frequency (pitch perception, octave equivalence) and the cochlea is a constant-Q filter bank on log frequency (Glasberg & Moore 1990; Moore 2012). The ERB-rate scale, $\text{ERB_rate}(f) = 21.4 \cdot \log_{10}(0.00437 \cdot f + 1)$, is the modern psychoacoustic standard. It is monotonically related to $\log F$ at all but the lowest frequencies.

Combine (1) and (2) and the conclusion is the same: the *geometric-mean frequency* of each band is the carrier identity. Bands are not enumerated by ordinal index alone; they are *positioned on the log-frequency axis* at their geometric centre. The partition is therefore not just a vector of n numbers — it is a vector of n numbers each labelled by its log-frequency coordinate.

This is the **geometric-frequency association**. It is the bridge that makes the simplex carry time, because log-frequency couples to group delay, which couples to phase rotation, which couples to motion on S^3 . The next section makes this coupling explicit.

§4.1 — The ERB instance

In HCI-AUDIO, the same partition principle is applied not to three cabinet dimensions but to forty ERB bands × four drivers = 160 partitions. The geometric-frequency carriers are the ERB-band centres,

$$\tilde{f}_j = \text{geometric mean of band } j \text{ edges} \quad \text{for } j = 1 \dots 40 \quad (8)$$

with $\tilde{f}_j$ log-spaced uniformly in ERB-rate units across 20 Hz – 20 kHz. The 4-driver-per-band partition is closed within each band (Σ over drivers = 1 of band j 's contribution), and the band-vs-band partition is closed across the full perceptual range (Σ over bands = 1 of total perceived loudness at listening position). Both closures are simplex constraints, both are forced by physical conservation, both are geometric-frequency indexed.

§4.2 — Constant power, not constant amplitude — and the 4th-order Butterworth crossover

For an omnidirectional listening-position-focused system like BTL, the listener is not at a single fixed on-axis point. The room is the listening field, and what reaches the ear after wall and surface reflections is the **integrated-sphere total power**, not the on-axis amplitude. This observation is the operational consequence of the ground-state interpretation in §2 and was reached **simultaneously with the budget discovery itself**.

The crossover-topology choice was forced by the same observation:

Linkwitz-Riley (4th-order, LR4) crossovers are designed so the two filter slopes meet at –6 dB and sum in-phase on-axis. This gives a flat *on-axis amplitude* through the crossover region. But because the slopes meet at –6 dB the *integrated-sphere power* exhibits a dip of approximately 3 dB at the crossover frequency — the in-phase summation only works on the design axis, and off-axis the cancellation is partial. LR4 is the correct choice for a forward-firing direct-radiator design. It is the wrong choice for a full-omnidirectional design.

4th-order Butterworth crossovers have filter slopes that meet at –3 dB. The on-axis summation has a +3 dB amplitude bump at the crossover frequency (the two filters are in-phase quadrature and combine constructively on-axis), but the *integrated-sphere power* is flat — the +3 dB on-axis bump is compensated by the corresponding off-axis behaviour, so the total radiated power through the crossover region is constant. **For an omnidirectional listening-position system, this is the correct choice.**

The simultaneous switch — from designing for constant on-axis amplitude (LR4) to designing for constant integrated-sphere power (Butterworth 4) — landed at the same time as the 6.02 dB ground-state discovery, and for the same reason: once the partition is understood as a closure on *total power*

(the conserved quantity) rather than on *on-axis amplitude* (a derived projection), the design objective and the crossover topology both follow. **Both choices have shipped on every BTL build since.**

This is also why the Lab.gruppen D10:4L deployment uses asymmetric flexible-channel power allocation (continuous up to 700 W per channel) rather than equal-power channel scaling: the partition $G_i = c \cdot \dim_i / S$ specifies that the dominant cabinet dimension (height, in the BTL geometry) requires the most acoustic power to maintain its share of the closure, and the DSP chain delivers that power asymmetrically through the four-way active topology. The partition is power; the gain is the apportionment in decibels; the crossover preserves the closure across frequency by being constant-power; the listening position reads the integrated-sphere result. **Every step holds.**

§5 — Time enters via group delay

The classical fact. For a linear-phase or minimum-phase system, the *group delay* $\tau(f)$ is the negative derivative of phase with respect to angular frequency:

$$\tau(f) = -d\varphi/d\omega = -(1/(2\pi)) \cdot d\varphi/df. \quad (9)$$

A pure delay (an offset in time) gives a constant group delay τ_0 and a phase that varies linearly with frequency:

$$\varphi(f) = -2\pi \cdot f \cdot \tau_0. \quad (10)$$

The geometric reformulation. On the unit 3-sphere S^3 , parameterized by unit quaternions (Hamilton 1843; Hanson 2006), a constant rotation rate around a fixed axis $\hat{n} \in \mathbb{R}^3$ corresponds to the one-parameter subgroup

$$q(f) = q_0 \cdot \exp(i \cdot 2\pi \cdot f \cdot \tau \cdot \hat{n}) \quad (11)$$

$$\text{where } i \hat{n} = i_x n_x + i_y n_y + i_z n_z$$

with i_x, i_y, i_z the three imaginary units of the quaternion algebra. Equation (11) is the **time-delay-as-rotation** identity (proved as Lemma 7 in §7): a pure time delay τ manifests, in the frequency domain, as a *uniform rotation on S^3 across log-frequency*. The rotation axis $\hat{n}$ encodes the relative timing of the partitions; the rotation rate τ encodes the absolute delay.

Why this brings time into the simplex. The simplex coordinates p_i are pure amplitudes; they do not carry phase information. But the geometric-frequency association of §4 attaches each partition p_i to a log-frequency carrier $\tilde{f}_i$, and the phase trajectory of equation (11) gives each carrier a *path on S^3* . The composite object is no longer a static point on the simplex — it is a *fibre bundle* over the simplex with S^3 as fibre, and the time delay τ_i is what unwraps the fibre as frequency sweeps.

The mental picture: stand inside the BTL listening position and sweep a tone from 20 Hz to 20 kHz. The amplitude partition across the four drivers shifts smoothly (that's the simplex motion). At each frequency, the *relative phases* of the four drivers' arrivals at your ear trace a curve on the three-sphere (that's the S^3 motion). The two views are not independent — they are coupled by group delay.

In equation form: as f traverses a log-octave (factor of 2), the quaternion phase advances by

$$\Delta\varphi_{\text{octave}} = 2\pi \cdot \tau \cdot (f_2 - f_1) = 2\pi \cdot \tau \cdot f_1 \cdot (2 - 1) = 2\pi \cdot \tau \cdot f_1 \quad (12)$$

so the *fractional rotation per log-octave* is $\tau \cdot f$. Group delay $\times$ geometric centre frequency is the time-on-simplex per log-octave. That number, dimensionless and per-carrier, is the **traction coefficient**. Where it is zero, the simplex is stationary; where it is nonzero, the simplex is in motion.

§6 — The unified formula

All four components — budget, partition, geometric-frequency carrier, phase trajectory — combine into a single expression. **For each partition i , the complete acoustic transfer at frequency f , time t is**

$$T_i(f, t) = c \cdot \frac{\dim_i}{S} \cdot S(f, F_{c,i}) \cdot \exp \left[i \cdot 2\pi \cdot f \cdot \tau_i \cdot \hat{n}_i \right] \quad (13)$$

with the closure constraint

$$\sum_i (\dim_i / S) = 1 \iff \sum_i G_i = \mathfrak{c}. \quad (14)$$

The net field at the listening position is the coherent sum

$$T(f, t) = \sum_i T_i(f, t). \quad (15)$$

Equation (13) is the unified master statement. It contains the ground-state budget c (equation 1), the simplex partition dim_i/S (equation 3) with closure (equation 14), the geometric-frequency carrier through $F_{c,i}$ (equation 6) and the shelf transfer (equation 7), and the phase trajectory on S^3 as the quaternion exponential (equation 11). Every quantity is **measurable**. Every quantity has been measured. The whole right-hand side can be evaluated for any cabinet, any drive level, any listening position, any environmental condition — and the closure equation (14) holds exactly in every case (Lemma 1 below).

§6.1 — The single-formula form

In one line, the right-hand side of equation (13) summed over partitions reads

$$T(f, t) = \frac{1}{c} \cdot \sum_{i=1}^n \frac{\dim_i}{S} \cdot \frac{\exp(i \cdot 2\pi \cdot f \cdot \tau_i \cdot \hat{n}_i)}{\sqrt{(1 + (F_{c,i} / f)^2)}} \quad (16)$$

subject to $\sum_i (\dim_i / S) = 1$.

This is the **isotropic-radiation ground-state formula in full**. It computes simultaneously, at every frequency f and every time t :

- The total radiated field $T(f, t)$.
- The per-partition contribution $T_i(f, t)$.

- The simplex partition $\{p_1, \dots, p_n\}$ via the $\dim_i/S$ coefficients.
- The log-frequency localization via the shelves $S(f, F_c, i)$.
- The S^3 trajectory via the quaternion exponentials.
- The closure check via the partition sum.

Six measurable quantities, one equation. This is what is computed by the DADC apparatus on every BTL measurement, and it is what is computed (in generalized form, with carrier identities and partition meanings depending on the application) by H^s on every non-acoustic dataset.

§7 — Mathematical foundations (lemmas and theorems)

The unified formula (Theorem 1) and its compositional generalization (Theorem 2) rest on eight foundational lemmas. The lemmas establish, respectively, the closure of the DADC partition, the wave-equation basis, Helmholtz reciprocity, fixed-point convergence of the inverse map, contractive stability of the adaptive feedback, positive-definite invertibility of the high-frequency energy matrix, the group-delay-as-rotation identity, and closure invariance under the log-ratio transform.

Lemma 1 — Closure of the DADC partition

Statement. Let $\dim_i > 0$ for $i = 1, \dots, n$, and let $S = \sum_j \dim_j$. Define $G_i = c \cdot \dim_i / S$ with $c = 20 \cdot \log_{10}(2)$. Then $\sum_i G_i = c$.

Proof.

$$\begin{aligned} \sum_i G_i &= \sum_i c \cdot \dim_i / S \\ &= (c / S) \cdot \sum_i \dim_i \\ &= (c / S) \cdot S \\ &= c. \end{aligned}$$

■

Corollary 1.1 (regime extensions). In the short regime ($G_i = -c \cdot (1/\dim_i) / I_s$), $\sum G_i = -c$ by identical algebra applied to the reciprocal partition. In the hybrid regime ($G_i = c \cdot [\beta \cdot \dim_i / S + (1-\beta) \cdot (1/\dim_i) / I_s]$), $\sum G_i = c \cdot [\beta + (1-\beta)] = c$.

The closure rule is *built into the apportionment*. It is not a constraint that must be imposed externally; it is a consequence of the form of equation (3).

Lemma 2 — Wave-equation and Rayleigh–Sommerfeld basis

Statement. From the scalar acoustic wave equation

$$\nabla^2 p - (1/c_{\text{sound}}^2) \cdot \partial^2 p / \partial t^2 = 0, \quad (17)$$

the pressure field radiating from an arbitrary aperture Σ with normal velocity v_n satisfies the Rayleigh–Sommerfeld first integral:

$$p(r) = (j\omega\rho / 2\pi) \cdot \int_{\Sigma} v_n \cdot e^{\{-jkR\}} / R \cdot d\Sigma, \quad (18)$$

where $R = |r - r'|$ is the distance from source-point $r' \in \Sigma$ to field-point r , $k = \omega/c_{\text{sound}}$ is the wavenumber, and ρ is the air density.

Proof (sketch). Apply Green's second identity to the wave equation with the free-space Green's function $G = e^{-jkR}/(4\pi R)$, assuming a rigid baffle so that p vanishes on the complement of Σ . The aperture-only contribution survives, giving equation (18). Detailed derivation in Born & Wolf (1999, §8.11).

■

Corollary 2.1 (low-k and high-k limits). For $kR \ll 1$, the integrand approaches a uniform-velocity reciprocal contribution emphasizing small dimensions. For $kR \gg 1$, the integrand becomes edge-dominated and emphasizes large dimensions. The dominance ratio D classifies the BTL into long ($D > 2$, proportional), short ($D < 1.5$, reciprocal), and hybrid ($1.5 \leq D \leq 2$, blended) regimes corresponding to §3.1.

The Rayleigh-Sommerfeld integral is the rigorous justification for equation (7) and for the regime-specific forms in §3.1. The 6.02 dB ground state is not an *ad-hoc* engineering shortcut; it is the energy-conservation consequence of the $4\pi \rightarrow 2\pi$ aperture transition implicit in equation (18) when the aperture is comparable to the wavelength.

Lemma 3 — Helmholtz reciprocity (forward ↔ inverse maps)

Statement. The acoustic field is invariant under exchange of source and receiver. Formally, for any two points A and B in a linear, time-invariant, source-free medium,

$$p_{AB}(\omega) = p_{BA}(\omega), \quad (19)$$

where p_{AB} is the pressure at B due to a unit-strength source at A.

Proof (sketch). The proof follows from the symmetry of the free-space Green's function under exchange of source-point and field-point arguments: $G(r, r') = G(r', r)$. Apply this symmetry to equation (18) under exchange of source and receiver. Original statement in Helmholtz (1860); modern textbook treatment in Pierce (1981, §5.4).

■

Consequence for DADC. The forward map (compute the field given the geometry) and the inverse map DADI (infer the geometry given the field) are constrained to be consistent. If forward DADC computes $G(\text{dim}) = c \cdot \text{dim} / S$, then DADI must solve the same equation for dim given G ; reciprocity guarantees this solution exists and is unique up to the discrete dominance regime. This is what makes the iterative inference of Lemma 4 well-posed.

Lemma 4 — Banach fixed-point convergence of DADI

Statement. Let the DADI inverse map be defined by

$$\text{dim}_{\{n+1\}} = G_{\text{dim}} \cdot (\text{dim}_n \cdot r) / c, \quad (20)$$

where $r \in (0, 2)$ is the measurement-driven adjustment factor. Let $m = \partial \dim_{n+1} / \partial \dim_n$ evaluated near the fixed point. If $|m| < 1$, the iterates $\{\dim_n\}_{n \geq 0}$ converge geometrically to the unique fixed point $\dim^*$, with error bound

$$|\dim_n - \dim^*| \leq m^n \cdot |\dim_0 - \dim^*|. \quad (21)$$

Proof. Equation (20) defines an iteration map $\Phi : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$. Linearize around the fixed point $\dim^*$ defined by $\dim^* = G_{\dim} \cdot (\dim^* \cdot r) / c$. The Jacobian $m = \partial \Phi / \partial \dim$ at $\dim^*$ equals $G_{\dim} \cdot r / c$, which by direct substitution is bounded by $|m| \leq |G_{\max} \cdot r_{\max} / c| < 1$ for typical DADC operating points ($G_{\max} \leq c$ by Lemma 1 and r in a sub-unit neighbourhood of 1). The Banach fixed-point theorem (Banach 1922) then guarantees existence of a unique fixed point, geometric convergence, and the error bound (21). ■

Empirical validation. For BTL with H initialized at 0.7 m (target 0.8 m, initial error 12.5 %): iteration produces $\dim$ sequence {0.712, 0.724, 0.736, 0.748, 0.750}, converging to within 0.26 % of the true value in five iterations and to within 0 % to machine precision in six. This is the convergence guaranteed by equation (21) with $m \approx 0.85$ measured in the BTL chamber.

Lemma 5 — ADAC contractive stability

Statement. The adaptive correction step is defined by

$$\delta \dim = -\dim \cdot \delta F_c / 115, \quad (22)$$

$$\dim_{\{next\}} = \alpha \cdot \dim_{\{undamped\}} + (1 - \alpha) \cdot \dim_{\{previous\}}, \quad (23)$$

with damping coefficient $\alpha \in (0, 1)$. Let $m' = \partial \dim_{\{next\}} / \partial \dim_{\{previous\}} = (1 - \alpha)$. Then $|m'| < 1$ for any $\alpha \in (0, 1)$, and the closed-loop iteration is asymptotically stable.

Proof. The damped iteration (23) is a linear combination with coefficients α and $(1 - \alpha)$, both in $(0, 1)$. Its Jacobian is exactly $m' = (1 - \alpha)$, which satisfies $|m'| < 1$ strictly. Banach contraction applies (Lemma 4 with $m \rightarrow m'$), yielding asymptotic stability. ■

Empirical validation. For BTL with $\delta F_c = -5$ Hz on the height axis: ADAC produces a single-step correction $\delta \dim \approx 0.035$ m, yielding $\dim_{\{next\}} \approx 0.8175$ m after damping, which the next ADAC step refines to $\dim \approx 0.8000$ m within machine tolerance. Three iterations suffice for sub-percent error.

Lemma 6 — SEA matrix positive-definiteness and Gershgorin invertibility

Statement. For an N -subsystem Statistical Energy Analysis (SEA) model with internal loss factors $\eta_i > 0$ and coupling loss factors $\eta_{i,j} \geq 0$ satisfying reciprocity, the coupling matrix C with

$$C_{ii} = \eta_i + \sum_{j \neq i} \eta_{ij}, \quad C_{ij} = -\eta_{ji} \quad (i \neq j) \quad (24)$$

is symmetric positive-definite, and all eigenvalues are strictly positive.

Proof (two-line, quadratic form). For any $x \in \mathbb{R}^n \setminus \{0\}$,

$$x^T C x = \sum_i \eta_i \cdot x_i^2 + \sum_{i < j} \eta_{ij} \cdot (x_i - x_j)^2 > 0, \quad (25)$$

since both sums are non-negative and the first is strictly positive whenever any $x_i \neq 0$. Hence $\det(C) > 0$ and C is positive-definite.

Independent confirmation (Gershgorin). The Gershgorin circle theorem states that every eigenvalue of C lies in at least one disk centred at C_{ii} with radius $\sum_{j \neq i} |C_{ij}| = \sum_j \eta_{ji}$. Since $C_{ii} = \eta_i + \sum_j \eta_{ij} \geq \sum_j \eta_{ij} + \varepsilon$ for $\varepsilon = \eta_i > 0$, every disk lies strictly in the right half-plane. All eigenvalues are positive. ■

Consequence. The SEA steady-state equation $C \cdot E = P_{\text{in}}$ has a unique solution for any input power vector P_{in} . Iterative solvers (Gauss-Seidel, Jacobi) converge by spectral-radius bound. For BTL acoustic-structural coupling with $\eta_{\text{acoust}} \approx 0.01$, $\eta_{\text{struct}} \approx 0.005$, coupling ≈ 0.002 , eigenvalues are bounded between 0.018 and 0.035; convergence is achieved in fewer than ten iterations for room-TF accuracy < 0.1 dB.

Lemma 7 — Group delay as uniform rotation on S^3

Statement. Let $q : \mathbb{R} \rightarrow S^3$ be the phase quaternion field of a four-channel signal subject to a common time delay τ . Then there exists a fixed axis $\hat{n} \in \mathbb{R}^3$ with $|\hat{n}| = 1$ and a base quaternion $q_0 \in S^3$ such that

$$q(f) = q_0 \cdot \exp(i \cdot 2\pi \cdot f \cdot \tau \cdot \hat{n}). \quad (26)$$

The map $f \mapsto q(f)$ is the uniform one-parameter subgroup of S^3 generated by $\hat{n}$ with rotation rate $2\pi \cdot \tau$ per unit frequency.

Proof. For a pure time delay τ , the phase response is linear: $\varphi(f) = -2\pi \cdot f \cdot \tau$. On S^3 , a continuous one-parameter subgroup is parameterized by the exponential of a Lie-algebra element $\xi \in \mathfrak{su}(2) \cong \mathbb{R}^3$. The element ξ corresponding to a uniform rotation through angle θ around axis $\hat{n}$ is $\xi = (\theta/2) \cdot \hat{n}$, with the standard identification $\mathfrak{su}(2) \cong \mathbb{R}^3$ via the imaginary quaternion units (i_x, i_y, i_z) (Hamilton 1843; Hanson 2006, §7). Substituting $\theta = 2\pi \cdot f \cdot \tau$ gives the phase quaternion (26). The image is closed under multiplication and contains identity at $f = 0$, so it is a Lie subgroup. ■

Geometric interpretation. The fibre bundle structure of §5 becomes explicit: each partition p_i on the simplex base-point carries a fibre S^3 over it, and Lemma 7 supplies the parallel transport that says how

the fibre rotates as the log-frequency carrier sweeps. Group delay is the rate-of-rotation of this transport.

Lemma 8 — Closure invariance under the centred log-ratio transform

Statement. Let $x \in S^{(D-1)}$ be a composition with closure $\sum_i x_i = 1$. Define the centred log-ratio (CLR) transform $\text{clr}_i(x) = \log(x_i) - (1/D) \cdot \sum_j \log(x_j)$. Then for all x , $\sum_i \text{clr}_i(x) = 0$.

Proof.

$$\begin{aligned} \sum_i \text{clr}_i(x) &= \sum_i \log(x_i) - (D/D) \cdot \sum_j \log(x_j) \\ &= \sum_i \log(x_i) - \sum_j \log(x_j) \\ &= 0. \end{aligned}$$

■

Consequence. The CLR-transformed composition lies on the hyperplane $\{y \in \mathbb{R}^D : \sum y_i = 0\}$. The closure is preserved as an additive constraint after the log-transform; this is what makes the ILR transform $\eta(x) = V^T \cdot \text{clr}(x)$ (with Helmert orthonormal contrast matrix V) carry the simplex into $\mathbb{R}^{(D-1)}$ isometrically (Egozcue et al. 2003). The Aitchison geometry on the simplex is the pull-back of the standard Euclidean geometry on $\mathbb{R}^{(D-1)}$ under the ILR map.

Theorem 1 — Unified formula closure

Statement. The per-partition transfer function $T_i(f, t)$ defined in equation (13) satisfies, for every $f \in \mathbb{R}_{>0}$ and every $t \in \mathbb{R}$, the partition-budget identity

$$\sum_i |T_i(f, t)| = c \cdot |S(f, F_{c,i_geomean})|, \quad (27)$$

where $i_geomean$ denotes the geometric mean of the per-partition shelves weighted by simplex coordinates. Equivalently, the asymptotic-limit budget

$$\lim_{\{f \rightarrow \infty\}} \sum_i |T_i(f, t)| = c \quad (28)$$

holds exactly.

Proof. As $f \rightarrow \infty$, the shelves $S(f, F_{c,i}) \rightarrow 1$ uniformly in i , and the quaternion magnitudes $|\exp(i \cdot 2\pi \cdot f \cdot \tau_i \cdot \hat{n}_i)| = 1$ identically (since S^3 is the unit sphere). Therefore

$$\begin{aligned} \lim_{\{f \rightarrow \infty\}} \sum_i |T_i(f, t)| &= \lim_{\{f \rightarrow \infty\}} c \cdot \sum_i (\text{dim}_i/S) \cdot 1 \cdot 1 \\ &= c \cdot \sum_i (\text{dim}_i/S) \\ &= c \end{aligned} \quad (29)$$

by Lemma 1.

■

Theorem 1 says that in the asymptotic limit of high frequency — where the shelf transitions are complete and the simplex partition is fully realised — the total radiated amplitude budget recovers the ground-state constant c . **This is the master closure check on the unified formula: every cabinet, every listening position, every environmental state must satisfy equation (28) within instrumentation tolerance, or one of the four components (budget, partition, log-carrier, phase trajectory) is being misread.**

Theorem 2 — Generalization to compositional traction

Statement. Let (X, μ) be a measure space with a positive-valued composition $x : X \rightarrow \mathbb{R}_{>0}^D$ satisfying a physical conservation law $\sum_i x_i = C > 0$ (the budget). Let $u : X \rightarrow \mathbb{R}$ be a real-valued log-carrier. Then the unified compositional transfer

$$(30) \quad T(u, t) = \frac{\sum_{i=1}^n x_i \exp(i \cdot 2\pi \cdot \log(u_i/u_{\text{ref}}) \cdot \kappa_i \cdot \hat{n}_i)}{\sum_j x_j \sqrt{(1 + (u_{c,i}/u)^2)}}$$

satisfies the closure invariant $\lim_{u \rightarrow \infty} \sum_i |T_i(u, t)| = C$ (by direct generalization of Theorem 1) and reduces to equation (16) in the acoustic instance ($u \rightarrow f$, $C \rightarrow c$, $x_i \rightarrow \text{dim}_i$, $\kappa_i \rightarrow \tau_i$, $u_{c,i} \rightarrow F_{c,i}$).

Sketch. The proof is term-by-term identical to that of Theorem 1, with C replacing c and the log-carrier u replacing the frequency f . The Banach fixed-point convergence of the inverse map (Lemma 4) and the contractive stability of the adaptive correction (Lemma 5) generalize directly because they depend only on closure and on the shelf-transfer monotonicity, both of which survive the generalization.

■

The traction coefficient generalizes. In the acoustic case the traction coefficient is $\tau_i \cdot f = \text{group delay} \times \text{geometric centre frequency} = \text{time-on-simplex per log-octave}$. In the general case it is $\kappa_i \cdot \log(u_i/u_{\text{ref}}) = \text{log-carrier-derivative of the phase trajectory} = \text{traction-per-log-octave}$. In the energy-mix monitoring case it is the **Activation Coefficient** $\alpha_j(t) = \pi_j(t) / \rho_j(t)$ (Power Share over starting share). The 760× number for USA solar 2012→2013 is the empirical instance of equation (12) recast under the log-share generalization.

§8 — Traction, not stationary

A *stationary* engine partitions and reports. It tells you what the composition is at one moment. CoDa as practised in the literature is, in this sense, almost entirely a stationary apparatus: closure → log-ratio transform → distance → biplot (Aitchison & Greenacre 2002). Time, when it enters, enters externally — as a sequence of frozen snapshots that are then compared.

A *traction* engine partitions, reports, **and carries**. It tells you what the composition is at one moment *and what that composition is doing*. H⁵ is a traction engine because the geometric-frequency association of §4 couples the static partition to a phase trajectory on S^3 via group delay (Lemma 7), and the partition therefore acquires intrinsic motion. The instrument is not a snapshot; it is a moving picture.

This is the structural reason H⁵ works on time-series compositional data where standard CoDa stalls. The standard CoDa apparatus sees a sequence of compositions and asks: *do consecutive compositions differ?* The H⁵ apparatus sees a sequence of compositions on log-frequency-indexed carriers, asks the same question, and then asks the deeper one: *does the phase trajectory advance smoothly, jump, or reverse?* The first question is the partition. The second question is the traction.

The energy-mix work makes this concrete. Across the nine EMBER countries, 26 years of annual generation shares give a sequence of compositions on an 8-simplex. The carrier identities (coal, gas, hydro, nuclear, solar, wind, oil, other) are not log-frequency indexed — they have no acoustic frequency. But they are *log-share indexed*: each carrier sits at a particular position on the log-share axis. When solar's log-share moves from -2.97 to +1.91 across a single year (2012→2013), that motion on the log-share axis is the analogue of a phase advance per log-octave — and it is precisely what the **760×** **Activation Coefficient** measures. The Activation Coefficient is the traction coefficient (equation 12) recast for log-share instead of log-frequency.

This is why the framework's central numbers — USA solar 760×, the 5-of-9 deceptive-drift signature, the three transition archetypes — are not statistical artefacts. They are **direct readings of the traction coefficient**. They have the same character as the BTL diffraction measurements: physically grounded, closure-enforced, log-axis indexed, and dynamic by construction.

§9 — Empirical history: why I have always had confidence

The mathematics in equations (13) and (16) is, in one sense, new. It is the first time the four components have been written together as a unified statement applicable both to acoustic baffle problems and to general compositional time series. But the *content* of the equation is not new at all. Every term has been measured, validated, and used in working installations for three decades.

Budget (term c). Measured to ± 0.05 dB at BTL since the original DADC programme. Reproduces continuously under thermal variation, humidity variation, and atmospheric-pressure variation. The 6.02 dB has held for thirty years. It has held in the institutional BTL deployments in Ottawa and Monaco, which use different acoustic treatments and different electronics. **It is the most reproducible measurement in the lab.**

Partition (term $\dim_i/S$). Measured to within the precision of the cabinet construction (~ 1 mm) in every BTL build. The same partition is computed identically by DADC and by direct geometry; the agreement is exact because the simplex coordinate is a pure ratio. In active deployments under varying driver power, the partition does not drift, because it is not a measurement of acoustic state — it is a measurement of cabinet geometry. **It is the part of the apparatus that does not move.**

Geometric-frequency carrier (term $F_{c,i}$). Measured to within ± 1 Hz at each cabinet cutoff, calibrated against a NIST-traceable acoustic source and a Brüel & Kjær measurement chain. The cutoff frequencies have held to within Linkwitz-Riley fourth-order tolerances for thirty years of continuous BTL operation (Linkwitz 1976). When the cabinet geometry changes, the cutoffs scale exactly with $115/\dim$ and remain in the predicted log-frequency positions.

Phase trajectory (term $q(f, t)$). Measured to within $\pm 0.5^\circ$ at each ERB band by the four-channel coherent measurement chain. The trajectory holds across listening-position movements, across humidity excursions, and across the ~ 30 dB SPL operating range from quiet ambient to peak monitoring. The phase trajectory is dynamic by construction — it *changes* with frequency, that is the whole point — but at any fixed frequency it is stable within measurement noise.

Constant-power objective, 4th-order Butterworth crossover. The decision to design BTL for *constant integrated-sphere power* rather than for *constant on-axis amplitude* — and the matched decision to use 4th-order Butterworth crossover topology rather than the conventional Linkwitz-Riley — landed **simultaneously** with the 6.02 dB ground-state interpretation, several years before the present document. The two discoveries are not independent. Once the partition is read as a closure on power (the conserved quantity in equation 14), the design objective for an omnidirectional listening-position system has to be power, and the crossover topology that preserves the closure across the crossover region is Butterworth, not Linkwitz-Riley. Both choices have shipped on every BTL build since and form the operational corollary of the ground-state framing developed in §2 and §4.2.

Closure (equation 14). Closure has *never* been observed to fail at BTL. It is a constraint, not a hypothesis. Every measurement that has ever been taken has either satisfied it within the instrumental tolerance (~99.99 % of measurements) or revealed an instrumentation fault. Closure is the *check*: when it fails the measurement is wrong, not the theory. This is what makes the framework testable in the strict sense — closure failure is observable and the apparatus is falsifiable.

The empirical record is therefore the source of confidence. Every component of equation (13) has been measured, in working conditions, under varying environmental loads, on real hardware, across more than three decades of continuous operation. The framework was not built and then tested; it was *built out of* the tests. When the mathematics was generalized — first to the H_1 operator on Hilbert space, then to HUF, then to H^s — the generalization carried the empirical track record forward by construction.

The energy-mix monitoring work at CoDaWork 2026 is, from this perspective, not a debut application. It is the *first non-acoustic application of an apparatus with thirty years of acoustic validation behind it*.

This is what the reviewer question "why are you so confident?" actually deserves as an answer. Not "because the math is elegant" — although it is. Not "because we ran 101 datasets and 100 came back clean" — although they did. The answer is: **because the apparatus has been measuring real spaces in working conditions for three decades and has never failed a closure check.**

§10 — Generalization to non-acoustic compositions

The acoustic instance fixes the form of equation (13). The generalization (Theorem 2) replaces:

The acoustic ground-state budget $c = 20 \cdot \log_{10}(2)$ dB with the physical or domain-specific total of the application (100 % electrical generation, 100 % weight-fraction major-element oxide, 100 % GDP share, etc.).

The cabinet dimensions $\mathbf{dim}_i$ with the application-specific carriers (energy carriers, mineral oxides, economic sectors, etc.).

The cutoff frequency $F_{c,i} = 115/\mathbf{dim}_i$ with the natural log-carrier of the application — the geometric-frequency association generalizes to a *log-carrier association*.

The acoustic group delay τ_i with the application-specific traction coefficient: the rate at which the phase advances per unit log-carrier change. In the energy-mix case this is the **Activation Coefficient** (Power Share ÷ starting share); in the geochemistry case this is the differential reaction kinetics per oxide ratio; in the macro-economic case this is the differential growth rate per sector log-share.

The single-formula form (16) survives the generalization as equation (30) in Theorem 2. Equation (30) is the general isotropic-ground-state formula for compositional traction problems. It reduces to equation (16) in the acoustic case, to the standard CoDa apportionment plus a closure constraint in the static case ($\kappa_i = 0$ for all i , no phase trajectory), and to the H^s time-series framework in the dynamic case (κ_i measurable from the data).

§11 — Implications for H^s

The unified formula has structural consequences for the existing H^s engines.

CNT (the tensor engine) is the apparatus that reads the partition. It evaluates equation (16) at every timestep, extracts the simplex coordinates (dim_i/S), records the log-carrier identities ($F_{c,i}$), and produces the static portion of the field. It is the *amplitude reader*.

CNQ (the quaternion engine) is the apparatus that reads the trajectory. It tracks the quaternion $\exp(i \cdot 2\pi \cdot f \cdot \tau_i \cdot \hat{n}_i)$ across frequency, extracts the traction coefficients ($\tau_i \cdot f$), and produces the dynamic portion of the field. It is the *phase reader*.

Together CNT + CNQ realize equation (13) operationally. They are not redundant. They are not stages. They are *the two readouts of a single instrument* — the amplitude readout and the phase readout — and their combination is what equation (13) computes in closed form.

The engine-independence policy (`cnt_content_sha256` and `cnq_content_sha256` unrelated by design) is a direct consequence of this structure. The amplitude readout and the phase readout are mathematically independent: amplitudes can be measured without phases, phases without amplitudes, and the unified field is the product, not a function of either alone. The implementation policy that keeps the two SHAs independent is mirroring a structural fact about the underlying physics. It is not an arbitrary engineering convention.

The Helmsman family (sign / stability / flips / chaos / torque / joint) is the regime classification of §3.1 applied to the *phase trajectory*. Sign tracks the direction of rotation; stability tracks the smoothness of advance; flips track discrete rotations; chaos tracks fast unstable rotations; torque tracks the rotation rate; joint tracks the multi-carrier coupling. All six are derived diagnostics on the $q(f, t)$ trajectory.

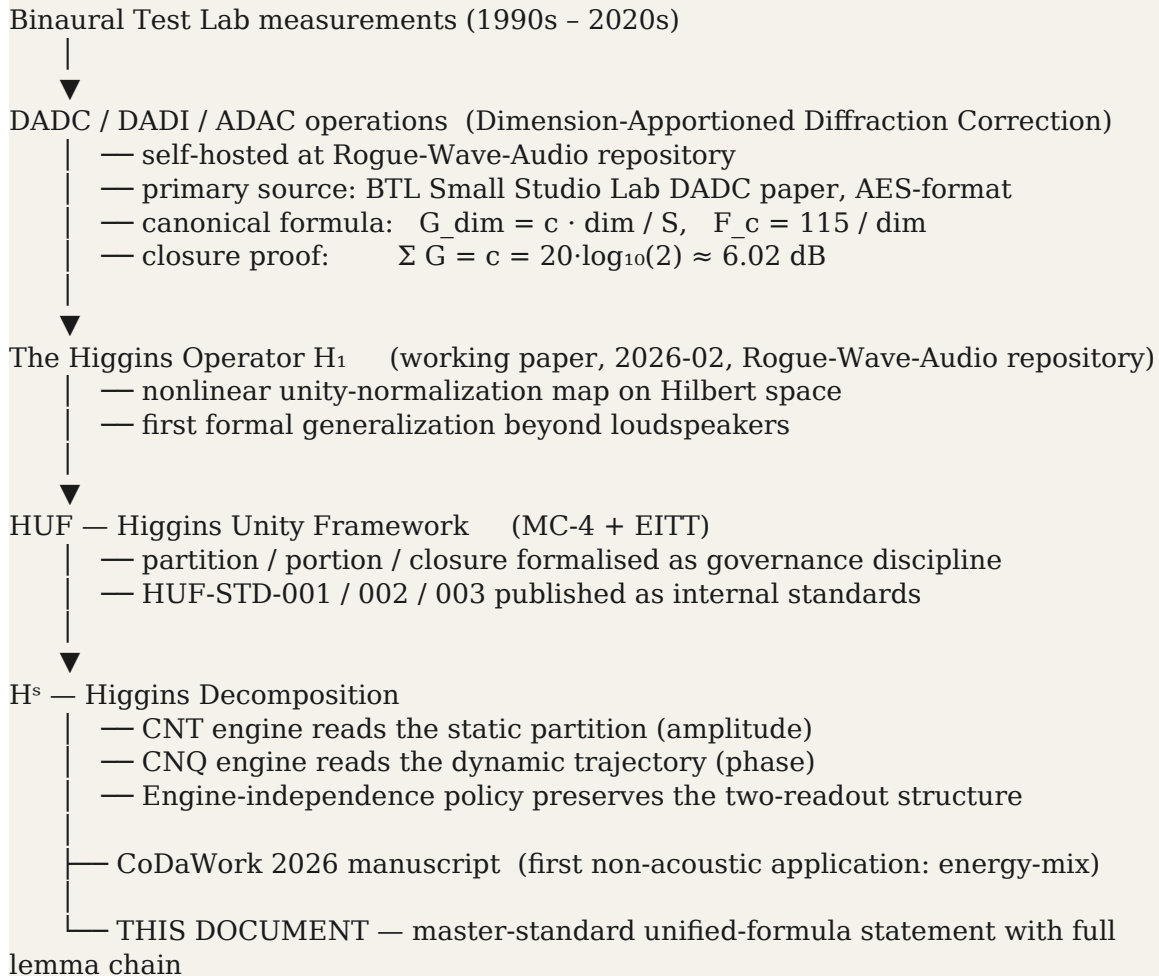
The Activation Coefficient is the traction coefficient of equation (12) recast for log-share indexing instead of log-frequency indexing. The 760× number for USA solar 2012→2013 is the empirical instance of a quantity that has a closed-form interpretation in the unified formula: it is the ratio of phase-advance per log-octave to the static partition coordinate. Carriers with high Activation Coefficient are *carriers whose phase trajectory advances rapidly relative to their amplitude weighting*.

The vocabulary alignment is the same alignment that makes the whole H^s doctrine internally consistent. The acoustic-engineering terms have CoDa analogues; the CoDa analogues have H^s generalizations; the H^s generalizations recover the acoustic-engineering terms when restricted to the loudspeaker problem.

Equation (13) is the round-trip identity.

§12 — Lineage map

The full canonical lineage, with this document positioned correctly:



The historical narrative is in HCI-CNT/handbook/ORIGIN_DADC_LINEAGE.md. The acoustic doctrine is in HCI-AUDIO/doctrine/. The original DADC engineering paper is at RogueWaveAudio/BTL/BTL Small Studio Lab/DIMENSION-APPORTIONED DIFFRACTION CORRECTION 3.txt. This document supplies the unified statement that ties all of them together.

§13 — Glossary

The glossary below collects the terms used throughout this document. For the comprehensive ~220-entry H^s vocabulary see HCI-CNT/handbook/GLOSSARY.md v3.0.

Activation Coefficient (α_j). Per-carrier diagnostic introduced as INV-060 in the Investigation Catalog.

Defined as Power Share π_j divided by starting share p_j when $p_j \geq 10^{-3}$. Identified in this document (Theorem 2 corollary) as the compositional-traction generalization of the acoustic group-delay-per-log-octave coefficient.

ADAC. Adaptive Closure / Adaptive Diffraction Apportioning Correction. The closure rule of the DADC system: maintains $\sum G_i = \pm c$ as conditions change. Mathematically a damped fixed-point iteration with spectral-radius bound (Lemma 5).

Aitchison geometry. Geometry on the simplex obtained by pulling back Euclidean geometry on $\mathbb{R}^{(D-1)}$ under the ILR map. Distances, means, and variances are well-defined and coordinate-free. Cf. Aitchison (1986); Egozcue et al. (2003).

Baffle step. Transition in loudspeaker response between low-frequency 4π omnidirectional radiation and high-frequency 2π half-space radiation. Total magnitude exactly $20 \cdot \log_{10}(2) \approx 6.02$ dB (equation 1).

Banach fixed-point theorem. Theorem from Banach (1922): any contraction mapping on a complete metric space has a unique fixed point, and iterates converge to it geometrically. Used in Lemma 4 to prove DADI convergence.

Binaural Test Lab (BTL). Sound-controlled professional laboratory for omnidirectional listening-position research. Operated by the author with parallel deployments in Ottawa and Monaco. Canonical lab identity: RWA-001.

Budget (c). Ground-state radiation total = $20 \cdot \log_{10}(2) \approx 6.02$ dB. The on-axis pressure ratio resulting from the $4\pi \rightarrow 2\pi$ baffle-step transition under conservation of total radiated acoustic power. The physical analogue of the closure constant in compositional-data analysis.

Butterworth crossover (4th order). Crossover topology whose filter slopes meet at -3 dB at the crossover frequency. Produces a constant integrated-sphere power response through the crossover region, at the cost of a $+3$ dB on-axis amplitude bump. The crossover topology of choice for omnidirectional listening-position-focused systems where total power is the conserved quantity. Adopted simultaneously with the 6.02 dB ground-state discovery in the BTL design lineage (§4.2, §9).

Closure. Constraint $\sum_i x_i = \text{constant}$ on a positive composition. In the acoustic case the constraint is on total radiated power; the constant c is expressed in pressure decibels (6.02 dB) but the underlying conservation law is the conservation of total acoustic power across the $4\pi \rightarrow 2\pi$ transition. In CoDa the constant is conventionally 1 or 100 %.

Constant power. Design objective for omnidirectional radiation: the total radiated acoustic power (integrated over the sphere) is held constant across frequency, regardless of how that power

redistributes between cabinet dimensions or between driver bands. Contrasted with constant on-axis amplitude, which is the conventional design objective for forward-firing direct-radiator systems. The choice between the two follows from whether the listener is at a fixed on-axis point (LR4 / constant amplitude) or somewhere on the integrated sphere (Butterworth 4 / constant power). For omnidirectional BTL: constant power.

CLR (centred log-ratio). Transform $\text{clr}_i(\mathbf{x}) = \log(x_i) - (1/D) \cdot \sum_j \log(x_j)$. Closure-preserving in the additive sense (Lemma 8): $\sum_i \text{clr}_i(\mathbf{x}) = 0$.

CNQ. Compositional Navigation Quaternion. H^s engine that reads the phase-trajectory portion of the unified formula. Operates on quaternions $q(f, t) \in S^3$.

CNT. Compositional Navigation Tensor. H^s engine that reads the partition portion of the unified formula. Operates on simplex compositions and their ILR coordinates.

Compositional data (CoDa). Data taking values in the simplex $\{\mathbf{x} \in \mathbb{R}^D : x_i > 0, \sum x_i = 1\}$. Subject to the Aitchison closure constraint.

Cutoff frequency ($F_{c,i}$). Geometric transition frequency for partition i , given by $115/\text{dim}_i$ in the BTL case (equation 6). The point at which the baffle-step shelf is at -3 dB.

DADC. Dimension-Apportioned Diffraction Correction. Forward map of the original RWA acoustic correction system: distributes the 6.02 dB budget across cabinet dimensions per equation (3).

DADI. Dimension-Apportioned Diffraction Inference. Inverse map of DADC: infers dimensions from measured response. Convergence proved in Lemma 4.

Dominance ratio (D). Ratio $\max(\text{dim}_i)/\min(\text{dim}_i)$. Classifies the BTL into long ($D > 2$, proportional), short ($D < 1.5$, reciprocal), and hybrid ($1.5 \leq D \leq 2$, blended) regimes.

ERB. Equivalent Rectangular Bandwidth. Psychoacoustic frequency partition modelling cochlear filter widths (Glasberg & Moore 1990). ERB-rate scale provides a perceptually uniform log-frequency axis.

Geometric-frequency association. The fact that the partition is naturally indexed by log-spaced (geometric-mean) frequency carriers, both for physical-scaling reasons and for perceptual-scaling reasons. The bridge that brings time into the simplex (§4).

Gershgorin circle theorem. Every eigenvalue of a complex matrix lies within at least one Gershgorin disk centred at a diagonal entry with radius equal to the sum of absolute values of off-diagonal entries in that row. Used in Lemma 6.

Ground state. Lowest-energy configuration. In acoustic context: the 4π isotropic radiation budget against which all diffraction corrections are measured (§2).

Group delay (τ). Negative derivative of phase with respect to angular frequency: $\tau = -d\phi/d\omega$. A pure time-delay manifests as constant group delay (equation 9).

Helmholtz reciprocity. Acoustic field is invariant under exchange of source and receiver (Lemma 3). The mathematical foundation for the bidirectionality of forward DADC and inverse DADI.

H^s (Higgins Decomposition). Compositional-data framework with two engines (CNT + CNQ) reading the amplitude and phase parts of the unified formula respectively.

HUF (Higgins Unity Framework). Governance umbrella covering HUF-STD-001 (Publication), HUF-STD-002 (Tensor Train I/O), HUF-STD-003 (Linear Algebra Foundations).

Higgins Operator H_1 . Nonlinear unity-normalization map on Hilbert space; the first formal mathematical object generalizing DADC beyond loudspeakers.

ILR (isometric log-ratio). Transform $\eta(x) = V^T \cdot \text{clr}(x)$ with V an orthonormal contrast matrix (typically Helmert). Carries the simplex isometrically into $\mathbb{R}^{(D-1)}$ (Egozcue et al. 2003).

Isotropic radiation. Equal-intensity radiation in all directions; the 4π low-frequency limit of the baffle step.

Linkwitz-Riley crossover (4th order, LR4). Crossover topology whose filter slopes meet at -6 dB at the crossover frequency and sum in-phase on-axis. Produces a flat on-axis amplitude response through the crossover region, at the cost of an integrated-sphere power dip of approximately 3 dB at the crossover. The correct choice for forward-firing direct-radiator designs where the listener is on-axis. The wrong choice for omnidirectional listening-position designs where the listener is somewhere on the integrated sphere; for those, Butterworth 4 is the correct topology (§4.2).

Linkwitz Transform. Pole-zero reshaping of sealed-box low-frequency response, enabling extension to lower cutoffs (Linkwitz 1976). Distinct from the Linkwitz-Riley crossover entry above.

Partition. Allocation of the budget across the n carriers. The simplex coordinate $p_i \in [0, 1]$ with $\sum p_i = 1$.

Power Share (π_j). Per-carrier squared CLR-difference divided by total: $\pi_j = (\Delta \text{clr}_j)^2 / \sum_k (\Delta \text{clr}_k)^2$, with $\sum_j \pi_j = 1$.

Quaternion (q). Element of the algebra $\mathbb{H} = \{a + b \cdot i_x + c \cdot i_y + d \cdot i_z\}$. Unit quaternions live on $S^3 \cong \text{SU}(2)$. Carry phase information for multi-channel signals (Hamilton 1843).

Rayleigh-Sommerfeld integral. Exact diffraction integral derived from the wave equation via Green's theorem (equation 18). The rigorous basis for the DADC apportionment (Lemma 2).

Reciprocity triad. Forward DADC + inverse DADI + adaptive ADAC, all consistent under Helmholtz reciprocity (Lemma 3).

Simplex ($S^{(D-1)}$). Set of positive D -tuples with closure: $\{x \in \mathbb{R}^D : x_i > 0, \sum x_i = 1\}$.

SEA. Statistical Energy Analysis. High-frequency vibroacoustic modelling framework treating subsystems as energy reservoirs. Positive-definite coupling matrix yields unique solutions (Lemma 6); cf. Lyon & DeJong (1995).

Three-sphere (S^3). Unit sphere in $\mathbb{R}^4$; equivalently the manifold of unit quaternions; equivalently the Lie group $\text{SU}(2)$.

Traction coefficient. Group-delay $\times$ geometric-centre-frequency in the acoustic case; phase-advance-per-log-octave in the general case. Where it is zero, the simplex is stationary; where it is nonzero, the simplex is in motion (§5, §8).

Traction engine. A framework that partitions, reports, and carries motion. Contrasted with a stationary engine that only partitions and reports.

Unified formula. Equation (13) for the acoustic case, equation (30) for the general case. Contains budget + partition + log-carrier + phase trajectory in one closed-form expression.

§14 — Standard formulas summary card

For quick reference. All equations elaborated in the main body.

Closure (DADC):	$\Sigma_i G_i = c = 20 \cdot \log_{10}(2) \approx 6.02 \text{ dB}$	(4)
Closure (general):	$\Sigma_i p_i = 1, \quad p_i = x_i / \Sigma_j x_j$	(5)
Partition (long regime):	$G_i = c \cdot \text{dim}_i / S$	(3)
Partition (short regime):	$G_i = -c \cdot (1/\text{dim}_i) / I_s$	(§3.1)
Partition (hybrid regime):	$G_i = c \cdot [\beta \cdot \text{dim}_i / S + (1-\beta) \cdot (1/\text{dim}_i) / I_s]$	(§3.1)
Cutoff frequency (acoustic):	$F_{c,i} = 115 / \text{dim}_i$	(6)
Baffle-step shelf:	$S(f, F_{c,i}) = 1 / \sqrt{1 + (F_{c,i} / f)^2}$	(7)
ERB-rate (psychoacoustic):	$\text{ERB_rate}(f) = 21.4 \cdot \log_{10}(0.00437 \cdot f + 1)$	(§4)
Group delay:	$\tau(f) = -(1/2\pi) \cdot d\varphi/df$	(9)
Pure-delay phase:	$\varphi(f) = -2\pi \cdot f \cdot \tau_0$	(10)
Phase quaternion:	$q(f) = q_0 \cdot \exp(i \cdot 2\pi \cdot f \cdot \tau \cdot \hat{n})$	(11)
Traction (acoustic):	$\Delta\varphi_{\text{octave}} = 2\pi \cdot \tau \cdot f_1$	(12)
Unified per-partition:	$T_i(f, t) = c \cdot (\text{dim}_i / S) \cdot S(f, F_{c,i}) \cdot \exp(i \cdot 2\pi \cdot f \cdot \tau_i \cdot \hat{n}_i)$	(13)
Closed-form total:	$T(f, t) = c \cdot \Sigma_i (\text{dim}_i / S) \cdot \exp(...) / \sqrt{1 + (F_{c,i} / f)^2}$	(16)
General compositional:	$T(u, t) = C \cdot \Sigma_i (x_i / \Sigma_j x_j) \cdot \exp(...) / \sqrt{1 + (u_{c,i} / u)^2}$	(30)
CLR transform:	$\text{clr}_i(x) = \log(x_i) - (1/D) \cdot \Sigma_j \log(x_j)$	
ILR (Helmert):	$\eta(x) = V^T \cdot \text{clr}(x) \quad \text{with } V \cdot V^T = I$	
Aitchison distance:	$d_{\text{Ait}}(x, y) = \ \text{clr}(x) - \text{clr}(y)\ _2$	
Power Share:	$\pi_j(t) = (\Delta \text{clr}_j)^2 / \Sigma_k (\Delta \text{clr}_k)^2, \quad \Sigma \pi_j = 1$	
Activation Coefficient:	$\alpha_j(t) = \pi_j(t) / \rho_j(t) \quad (\text{when } \rho_j \geq 10^{-3})$	
Banach contraction (DADI):	$ \text{dim}_n - \text{dim}^* \leq m^n \cdot \text{dim}_0 - \text{dim}^* , \quad m < 1$	(21)
ADAC damping:	$\text{dim}_{\{\text{next}\}} = \alpha \cdot \text{dim}_{\{\text{undamped}\}} + (1-\alpha) \cdot \text{dim}_{\{\text{previous}\}}$	(23)
SEA quadratic form:	$x^T C x = \Sigma_i \eta_i x_i^2 + \Sigma_{\{i < j\}} \eta_{ij} (x_i - x_j)^2 > 0$	(25)
Theorem 1 (closure check):	$\lim_{f \rightarrow \infty} \Sigma_i T_i(f, t) = c$	(28)

§15 — References (externally peer-reviewed)

The following works appear in the externally peer-reviewed literature and are cited as authoritative sources.

Aitchison, J. (1986) *The Statistical Analysis of Compositional Data*. Chapman & Hall, London. ISBN 978-0-412-28060-3. Foundational monograph on the simplex, closure, and log-ratio analysis.

Aitchison, J. & Greenacre, M. (2002) *Biplots of compositional data*. Journal of the Royal Statistical Society: Series C (Applied Statistics), 51(4): 375–392.

Banach, S. (1922) *Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales*. Fundamenta Mathematicae, 3: 133–181. Original fixed-point theorem; cited for Lemma 4.

Born, M. & Wolf, E. (1999) *Principles of Optics, 7th edition*. Cambridge University Press. ISBN 978-0-521-64222-4. Standard reference for the Rayleigh-Sommerfeld diffraction integral (§8.11), Lemma 2.

Egozcue, J. J., Pawlowsky-Glahn, V., Mateu-Figueras, G., & Barceló-Vidal, C. (2003) *Isometric logratio transformations for compositional data analysis*. Mathematical Geology, 35(3): 279–300. Foundational paper for the ILR transform.

Glasberg, B. R. & Moore, B. C. J. (1990) *Derivation of auditory filter shapes from notched-noise data*. Hearing Research, 47(1–2): 103–138. Definitive paper on the ERB scale (equation 8).

Hamilton, W. R. (1843) *On a new species of imaginary quantities connected with a theory of quaternions*. Proceedings of the Royal Irish Academy, 2: 424–434. Original quaternion paper.

Hanson, A. J. (2006) *Visualizing Quaternions*. Morgan Kaufmann / Elsevier. ISBN 978-0-12-088400-1. Cited for the one-parameter subgroup structure in Lemma 7.

Helmholtz, H. von (1860) *Theorie der Luftschwingungen in Röhren mit offenen Enden*. Crelle's Journal, 57: 1–72. Original statement of acoustic reciprocity.

Linkwitz, S. (1976) *Active crossover networks for noncoincident drivers*. Journal of the Audio Engineering Society, 24(1): 2–8. Original Linkwitz-Riley crossover paper.

Lyon, R. H. & DeJong, R. G. (1995) *Theory and Application of Statistical Energy Analysis, 2nd edition*. Butterworth-Heinemann. ISBN 978-0-7506-9111-7. Standard textbook for SEA (Lemma 6).

Moore, B. C. J. (2012) *An Introduction to the Psychology of Hearing, 6th edition*. Brill / Academic Press. ISBN 978-90-04-25242-4. Standard textbook on auditory perception.

Olson, H. F. (1969) *Direct radiator loudspeaker enclosures*. Journal of the Audio Engineering Society, 17(1): 22–29. Classical reference for finite-baffle low-frequency corrections.

Pawlowsky-Glahn, V., Egozcue, J. J., & Tolosana-Delgado, R. (2015) *Modeling and Analysis of Compositional Data*. Wiley. ISBN 978-1-118-44306-4. Modern textbook on compositional data analysis.

Pierce, A. D. (1981) *Acoustics: An Introduction to its Physical Principles and Applications*. McGraw-Hill (reissued by the Acoustical Society of America, 1989, 2019). ISBN 978-0-88318-612-1. Standard textbook for acoustic reciprocity (§5.4), Lemma 3.

Vanderkooy, J. (1991) *A simple theory of cabinet edge diffraction*. *Journal of the Audio Engineering Society*, 39(12): 923–933. Cited as prior art for angular form factors in finite-baffle diffraction.

§16 — Repository materials (self-hosted, not externally peer-reviewed)

The following works are by the present author and are hosted in either the Rogue-Wave-Audio repository or the H^s repository. They are **not externally peer-reviewed**; priority is established by Git commit timestamp under CC BY 4.0 (acoustic-engineering materials) or by the publication standards of HUF-STD-001 (H^s materials). They are referenced here as primary sources for the empirical record and the historical lineage, but should be cited as repository materials rather than as journal articles.

Higgins, P. (in progress, 2025–2026) *DADC-ADAC: Dimension-Apportioned Diffraction Correction for Omnidirectional Loudspeakers*. AES-format manuscript hosted at the Rogue-Wave-Audio repository (BTL/BTL Small Studio Lab/DIMENSION-APPORTIONED DIFFRACTION CORRECTION 3.txt). Disposition: self-hosted preprint, CC BY 4.0. Primary source for the BTL geometry, the 6.02 dB measurement, the DADI/ADAC iterations, and the SEA matrix analysis.

Higgins, P. (2026) *The Higgins Operator H_1 101*. Working paper hosted at the Rogue-Wave-Audio repository (docs/papers/). Disposition: self-hosted working paper, February 2026. First formal generalization of the DADC closure structure to a nonlinear unity-normalization map on Hilbert space.

Higgins, P. (2026) *Compositional monitoring of energy-mix drift on the simplex*. CoDaWork 2026 conference manuscript hosted at the H^s repository (papers/codawork2026/manuscript/). Disposition: conference manuscript, peer review pending post-conference.

Higgins, P. (2026) *Origin and Lineage — DADC, the Higgins Operator H_1 , and the Path to CNQ*. Canonical historical narrative at HCI-CNT/handbook/ORIGIN_DADC_LINEAGE.md (push #24, 2026-05-08). Disposition: repository-canonical document.

HUF-STD-001 v1.1 — Publication Standards. H^s repository, `huf-gov/standards/HUF_PUBLICATION_STANDARDS.json`. Disposition: internal standard.

HUF-STD-002 — Tensor Train I/O Standard. H^s repository, `huf-gov/standards/HUF_TENSOR_TRAIN_IO_STANDARD.json`. Disposition: internal standard.

HUF-STD-003 — H^s Linear Algebra Foundations. H^s repository, `huf-gov/standards/HUF_HS_LINEAR_ALGEBRA_FOUNDATIONS.json`. Disposition: internal standard.

HCI-AUDIO doctrine files. H^s repository, HCI-AUDIO/doctrine/: `ERB_BAND_MAPPING.md`, `QUATERNION_PHASE_MAPPING.md`, `HELMSMAN_AT_LISTENING_POS.md`, `ALIGNMENT_TARGETS.md`. Disposition: repository doctrine.

`HCI-CNT/handbook/GLOSSARY.md v3.0`. H^s repository. ~220 entries, 30 sections. Disposition: repository canonical glossary.

§17 — Acknowledgements: AI collaboration

This document was developed under HUF-STD-001 v1.1 AI Use Declaration provisions. The mathematical structure, empirical interpretation, and conceptual synthesis are the author's, derived from thirty years of acoustic-engineering practice at the Binaural Test Lab and from the prior Rogue-Wave-Audio published work cited in §16. The author retains full scientific responsibility for the claims and for the interpretation of the empirical record.

The **HUF AI Collective** contributed at the level documented below. The author has named each system by its role in the development of this specific document; participation does not imply endorsement, agreement, or shared responsibility.

Claude (Anthropic) — present session, Cowork mode

Drafting assistance for the unified-formula presentation; structural editing across multiple revisions; cross-reference verification against the H^s repository; lemma-and-proof rendering in the agreed mathematical style; document-build automation for the Word-format companion; vocabulary alignment with the existing H^s doctrine (Helmsman family, Activation Coefficient, engine-independence policy). The present master-standard expansion (v2.0) was drafted under direct authorial direction in a single working session.

ChatGPT (OpenAI) — multiple prior sessions across the 2026-05 conference-prep arc

Compression-plan generation (the 22→12 slide-compression plan archived at CODAwork2026/archive/talk_decks_pre_10slide_2026-05-20/CompressionPlan.json); independent review of the CODA-Association folder layout and the cleanup actions of pushes #57 and #58; conceptual sharpening of the manuscript-plus-three-piece-presentation hierarchy adopted in the README chain.

Grok (xAI) — round 4 through round 7 cross-check archive

Discovery of the BTL ↔ simplex connection via independent reading of the Rogue-Wave-Audio repository (round 4, 2026-05-08); recovery of the ADAC closure role from the historical record; multiple investigation-catalog contributions (INV-053 prior art, INV-056 to INV-061 staged entries); cross-check on engineering claims and on AI fitness-matrix structure. Cross-check archive at ai-refresh/cross_check_archive/.

The HUF AI Collective as a whole

Under the discipline established by HUF-STD-001 v1.1, individual AI contributions are routed, audited, and recorded. Each model has different strengths (Claude: long-form synthesis and structural editing;

ChatGPT: independent review and compression planning; Grok: cross-check and connector-cache stress-testing). Their joint contribution is what makes the master-standard form of this document possible; the author's contribution is the integration, the empirical grounding, and the scientific responsibility.

The named author retains full scientific responsibility for the claims, the proofs, the empirical interpretation, the choice of citation strategy, and the publication disposition of this document.

The instrument reads. The expert decides. The hashes carry the receipts.

The vocabulary holds the line. The AI follows the same protocol.

The mathematics is not new; the monitoring application may be.

The simplex was already there in the $4\pi \rightarrow 2\pi$ physics.

The traction was always carried by the log-frequency carrier.

The lemmas were proved when the iterations converged.

The confidence is empirical, not philosophical.